# VO XUAN NHI TRAN

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## ABOUT ME

Third-year Software Engineering Honours student with hands-on full-stack development experience in React, FastAPI, and AWS. Through recent technical projects, I engineered an AI-powered pipeline that processed audio synchronization with 0.00s timeline drift, and built a deterministic visa screener evaluated on 50 real-world job ads. I bring strong debugging skills and cloud deployment expertise to help tech teams build robust, scalable, and explainable software solutions.

## EDUCATION

**University of Sydney** July 2024 – Present | Sydney, Australia

**Bachelor of Engineering Honours (Software Engineering) - Dalyell Scholar - Vice Chancellor's International Scholarship**

- **WAM:** 77.75/100 (Distinction average) | **Expected Graduation:** 2028
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Agile Software Development, Security Engineering, Software Construction and Design.

## PROJECTS

**RAG Document Assistant** GitHub | Live Demo

*Technologies: Python, FastAPI, React, PostgreSQL/pgvector, Docker, AWS*

- Built a full-stack RAG app (React, FastAPI, PostgreSQL/pgvector, Gemini API) where users upload PDFs and get answers grounded in their documents; deployed on AWS EC2 with Nginx and CI/CD, debugging CORS and HTTPS mixed-content issues along the way.
- Wrote 13 pytest tests with mocked LLM calls and an isolated test database, including cross-session isolation tests that verify one user's documents are never retrievable by another.

**JobCompass (Futura Remix Hackathon 2026)** GitHub | Live Demo

*Technologies: Next.js, TypeScript, Gemini API, Upstash Redis, PostHog, Tailwind CSS, Vitest*

- Built an AI visa-eligibility screener for international students that outputs explainable APPLY / TAILOR / SKIP decisions backed by exact quotes from the job ad.
- Paired a deterministic rules engine with Gemini extraction, dropping any claim without direct evidence to prevent hallucination.
- Raised extraction accuracy from 60% on the hold-out set to 66.67% by rewriting the logic around clause structure (50 job ads: 35 dev, 15 hold-out).

**YouTube Video Dubber (Chinese to Vietnamese)** GitHub

*Technologies: Python, faster-whisper, Gemini API, ffmpeg*

- Automated a Chinese-to-Vietnamese dubbing pipeline (faster-whisper transcribe, Gemini translate, TTS, ffmpeg merge).
- Fixed a 35% audio overrun with a word-budget translation step based on measured TTS timings and silence trimming, reaching 0.00s drift across 13 blocks on a 72-second clip.
- Added caching and retry/backoff to work around Gemini free-tier quota limits.

**RenoPilot – Roof Replacement App | Capstone, team of 8**

*Technologies: Next.js, React, TypeScript, Payload CMS, PostgreSQL*

- Built the four-option landing page by extending the shared AppLanding component without breaking the other RenoPilot apps, then added persistent light/dark themes across the quote workflow (merged, client-accepted).
- Regression-tested the Property → Map → Materials → Quote flow and Undo feature (148 tests passing), reporting bugs to feature owners until fixed.
- Rotated through Tester Lead, Programmer Lead and Customer Liaison, presenting progress to the client.

## TECHNICAL SKILLS

- **Languages:** Python, Java, TypeScript/JavaScript, C, SQL, HTML/CSS
- **Frameworks:** React, Next.js, FastAPI, Django, Django REST Framework
- **Databases:** PostgreSQL, pgvector, SQLite
- **Cloud & DevOps:** Docker, AWS EC2, Nginx, Vercel, Render, Linux
- **Tools:** Git/GitHub, pytest, JUnit, Gemini API